STaRK: Benchmarking LLM Retrieval on Textual and Relational Knowledge Bases

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Joohyung Yun
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Overview of This Seminar

- Introduction
- Background
- Problem Definition
- Methodology of Benchmark Generation
  - Generated Semi-structured Knowledge Bases
  - Query-Ground Truth Entity Set Generation
- Analysis of Benchmark
- Experiments

Introduction

Semi-Structure Retrieval Benchmark

Description: Generate a benchmark for the task of *retrieving nodes via natural language* on semi-structured knowledge base.

Example use case: Information retrieval on e-commerce web search

- People express complex concepts that need both structured and unstructured information.

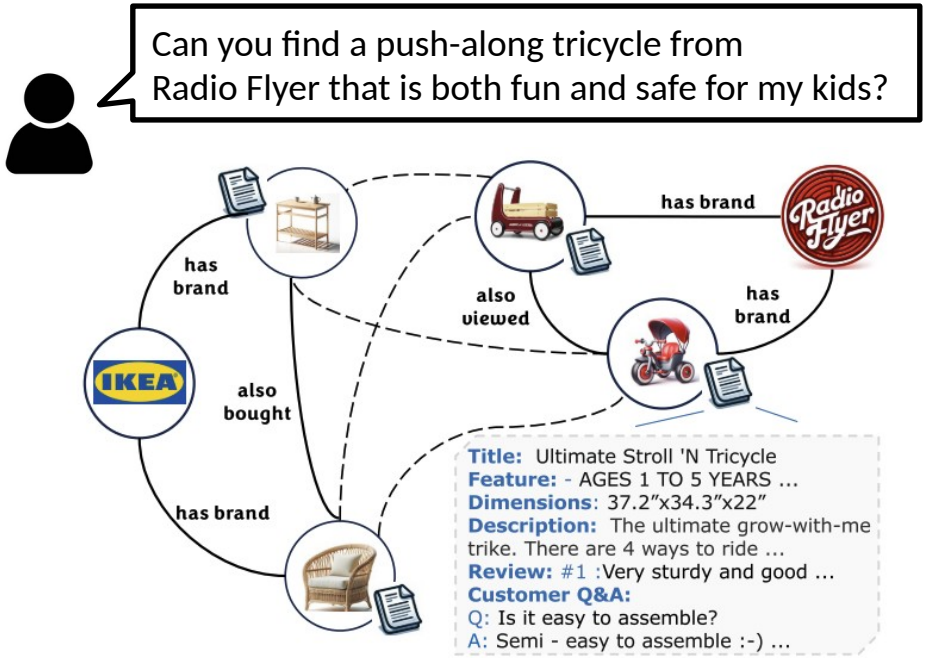


Figure 1. An example case of semi-structure retrieval

Motivation of Research

Problems of previous works

- Focused on either purely textual queries on unstructured knowledge or structured SQL or knowledge graph queries.
- Multi-hop reasoning benchmarks are either limited in the span of knowledge or lack in diverse textual properties and explicit relational information.
- LLM retrieval techniques remain an open question of how to effectively apply them to the specific challenge of retrieval from semi-structured knowledge bases.

Contributions

- **Natural-sounding queries on semi-structured knowledge:** the queries are crafted to incorporate rich relational information and complex textual properties.
- **Context-specific reasoning:** The queries entail reasoning capabilities specific to the context.
- **Diverse domains:** wide range of content and domains, reflecting the diverse nature of real-world information needs.

Background

Background

Semi-Structured Knowledge Base (SKB)

A semi-structured knowledge base corresponds to a *labeled property graph*.

Component in labeled property graph	Component in semi-structured knowledge base
Node	Entity
Node label	Entity type
Node properties and values	Textual information
Edge	Relation
Edge label	Relation type

Table 1. Relationship between the components of a labeled property graph and the components in SKB

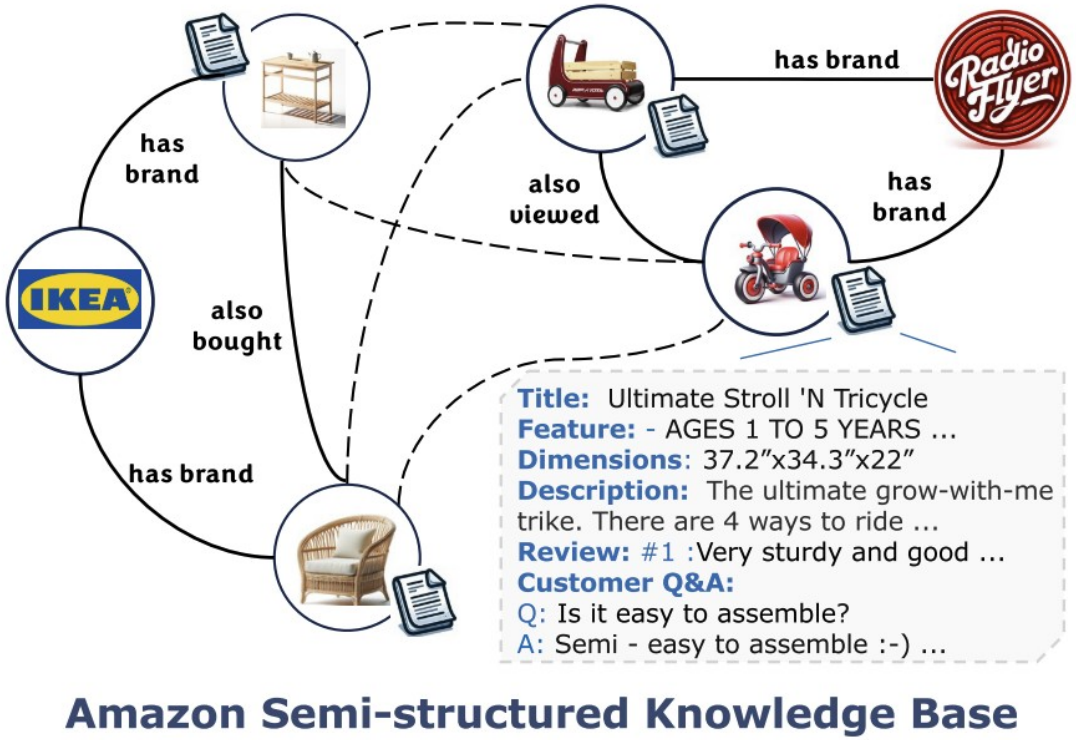


Figure 2. An example SKB

Background

Semi-Structured Knowledge Bases (SKB)

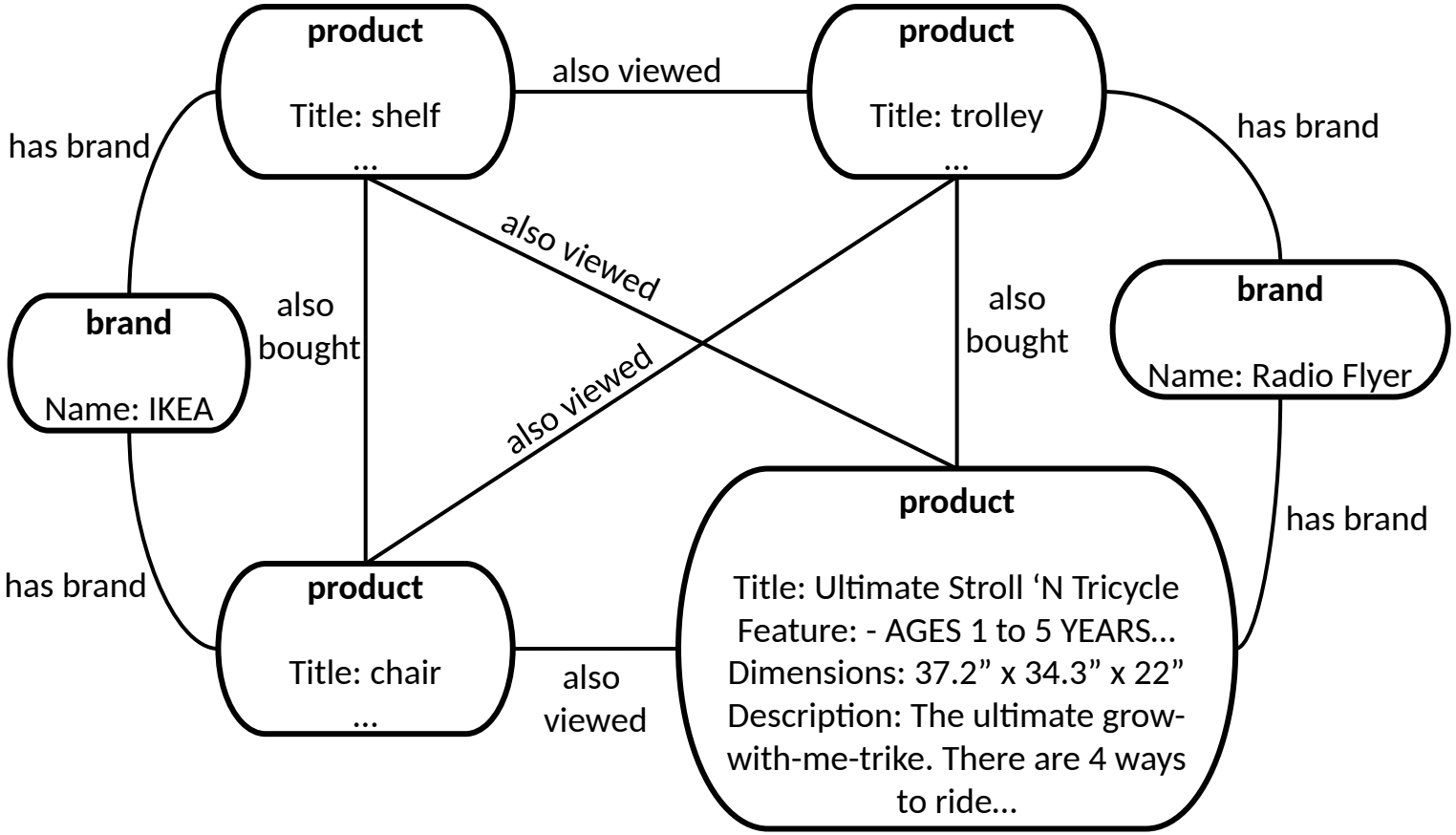


Figure 3. Representation of SKB in Figure 2. in a labeled knowledge graph

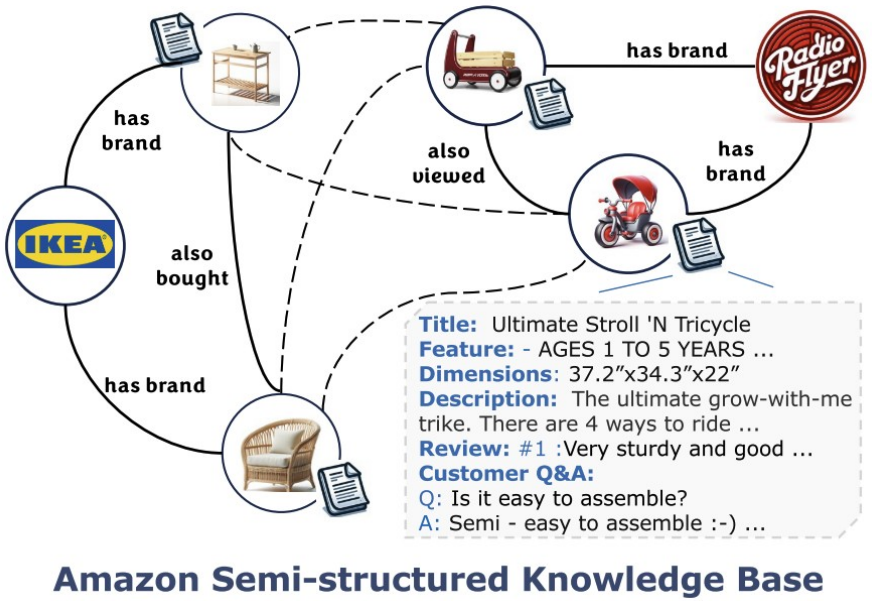


Figure 2. An example SKB

Problem Definition

Problem Definition

Retrieval Task on SKB

Description: Retrieve node entities over an SKB given a natural language query.

- Input: natural language query
- Input: SKB
- Output: a set of nodes within the SKB

	Example query	Title of ground truth node(s)
STARK-AMAZON	*Looking for durable Dart World brand dart flights that resist easy tearing. Any recommendations?*	<Amazon Standard Flights> <Dart World Broken Glass Flight> (12 more)
	What are recommended scuba diving weights for experienced divers that would fit well with my Gorilla PRO XL waterproof bag?	<Sea Pearls Vinyl Coated Lace Thru Weight>
STARK-MAG	*Search publications by Hao-Sheng Zeng on non-Markovian dynamics.*	<Distribution of non-Markovian intervals...> <Comparison between non-Markovian...>
	What are some nanofluid heat transfer research papers published by scholars from Philadelphia University?	<A Numerical Study on Convection Around A Suqare Cylinder using AL2O3-H2O Nanofluid>
STARK-PRIME	*Could you provide a list of investigational drugs that interact with genes or proteins active in the epididymal region?*	<(S)-3-phenyllactic Acid>, <Anisomycin>, <Puromycin>
	Search for diseases without known treatments and induce pruritus in pregnant women, potentially associated with Autoimmune.	<Intrahepatic Cholestasis>
	Please find pathways involving the POLR3D gene within nucleoplasm.	<RNA Polymerase III Chain Elongation>
	Which gene or protein associated with lichen amyloidosis can bind interleukin-31 to activate the PI3K/AKT and MAPK pathways?	<OSMR>, <IL31RA>

Table 2. Example pairs of query and answer node set for SKB retrieval task

Problem Definition

Benchmark for Retrieval Task on SKB

A benchmark for retrieval task on SKB is comprised as

- A semi-structured knowledge base
- A set of (query, ground-truth entity set) pairs

Benchmark Generation

Generated SKBs

- STaRK has generated 3 SKBs
- The statistics for each SKB are as follows:

	#entity types	#relation types	avg. degree	#entities	#relations	#tokens
STARK-AMAZON	2	3	3.0	1,032,407	3,886,603	592,067,882
STARK-MAG	4	4	10.6	1,872,968	19,919,698	212,602,571
STARK-PRIME	10	18	62.6	129,375	8,100,498	31,844,769

Table 3. Statistics for the 3 semi-structured knowledge bases of STaRK

Generated SKB : Example

MAG Semi-structured Knowledge Base

Generation steps:

1. Select papers shared between *obgn-MAG* and *obgn-papers100M*.
2. Extract subgraph of *obgn-MAG* based on the selected papers to earn relational structure.
3. Extract textual information such as titles and abstracts for papers from *obgn-papers100M*.
4. Augment the textual information by integrating details (e.g., venue, author and institution names) from the *Microsoft Academic Graph database*.

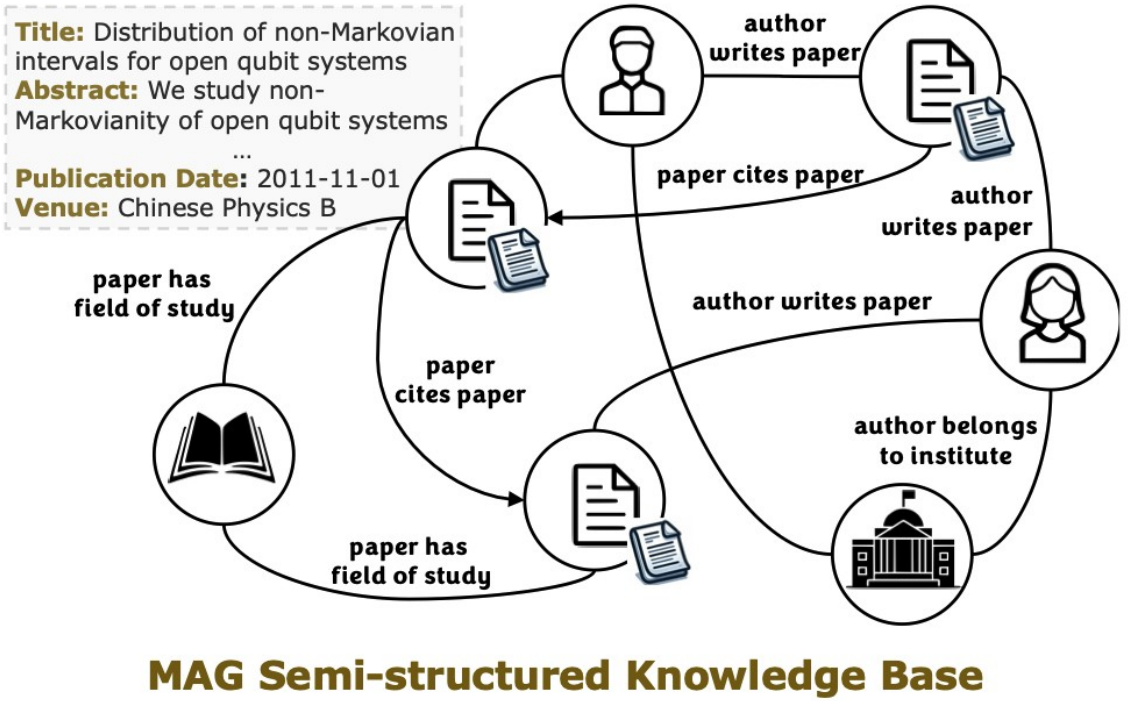


Figure 5. The MAG semi-structured knowledge base

Query-Ground Truth Entity Set Generation

Overall Idea

The process of generating a (query, ground truth entity set) pair from a SKB can be interpreted as

1. **Query generation:** Picking a subgraph of SKB
    1. Determine the nodes, node labels, relations types of a subgraph.
    2. Determine the properties of a subgraph.
2. **Ground truth entity set generation:** Filtering entities that match the subgraph of (1)

Query-Ground Truth Entity Set Generation

Overview

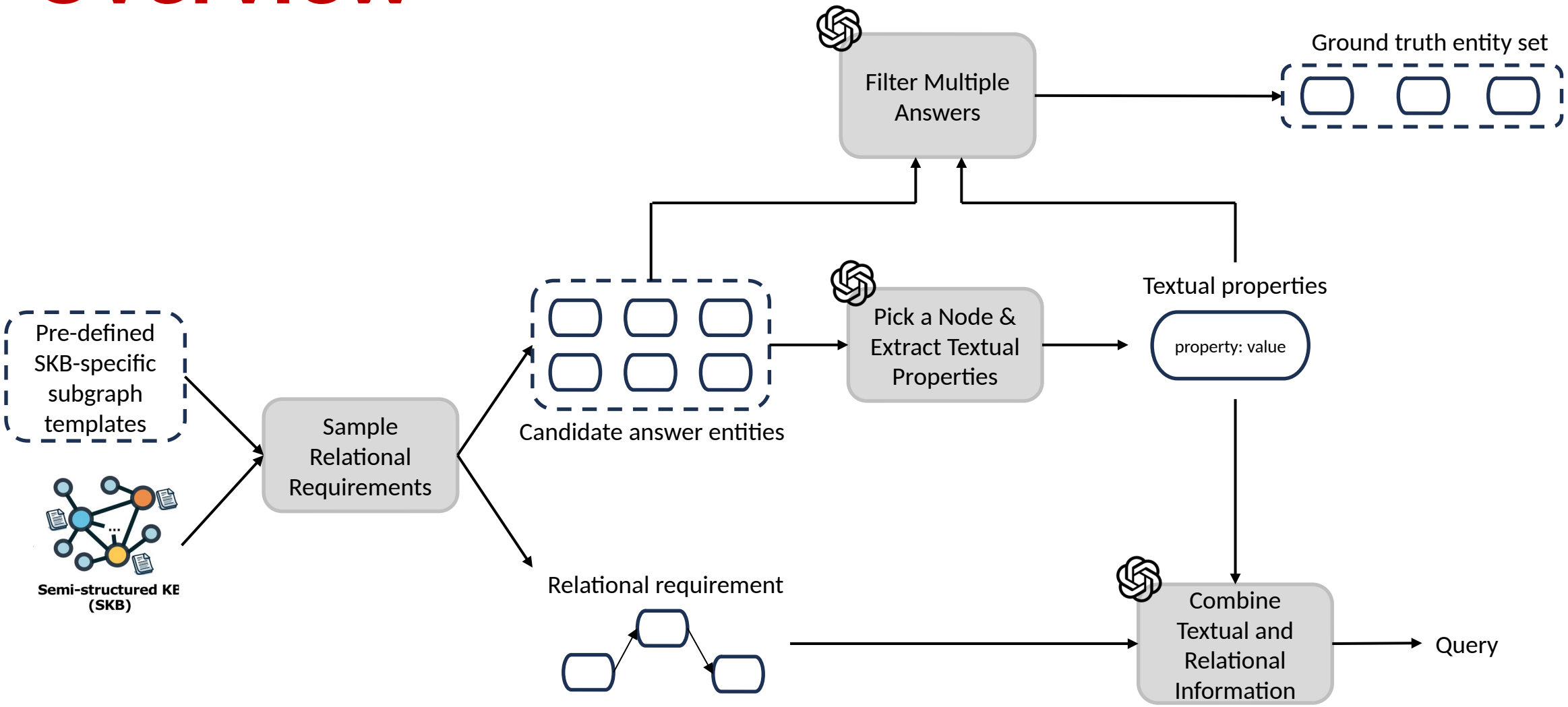
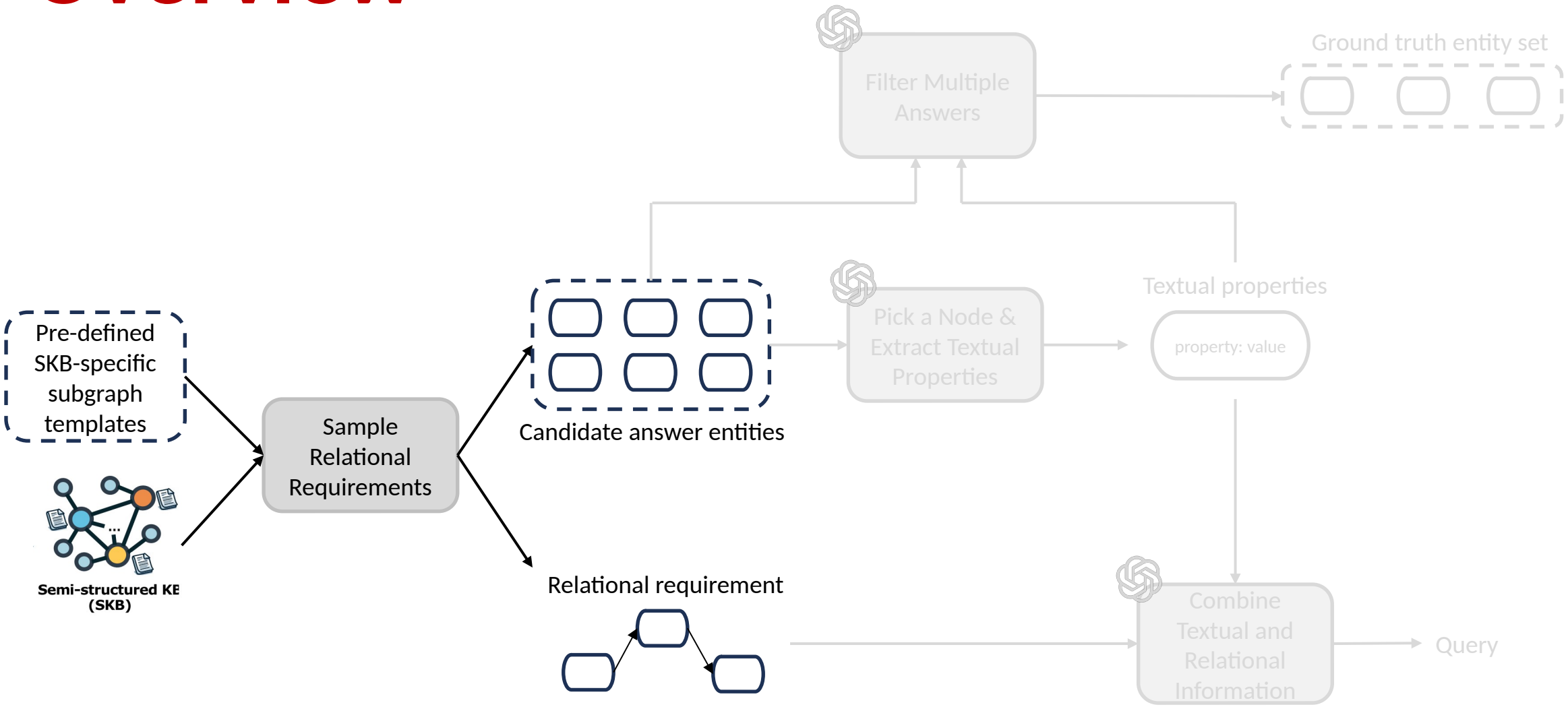


Figure 4. Overall flow of a single query – ground truth entity set pair generation.

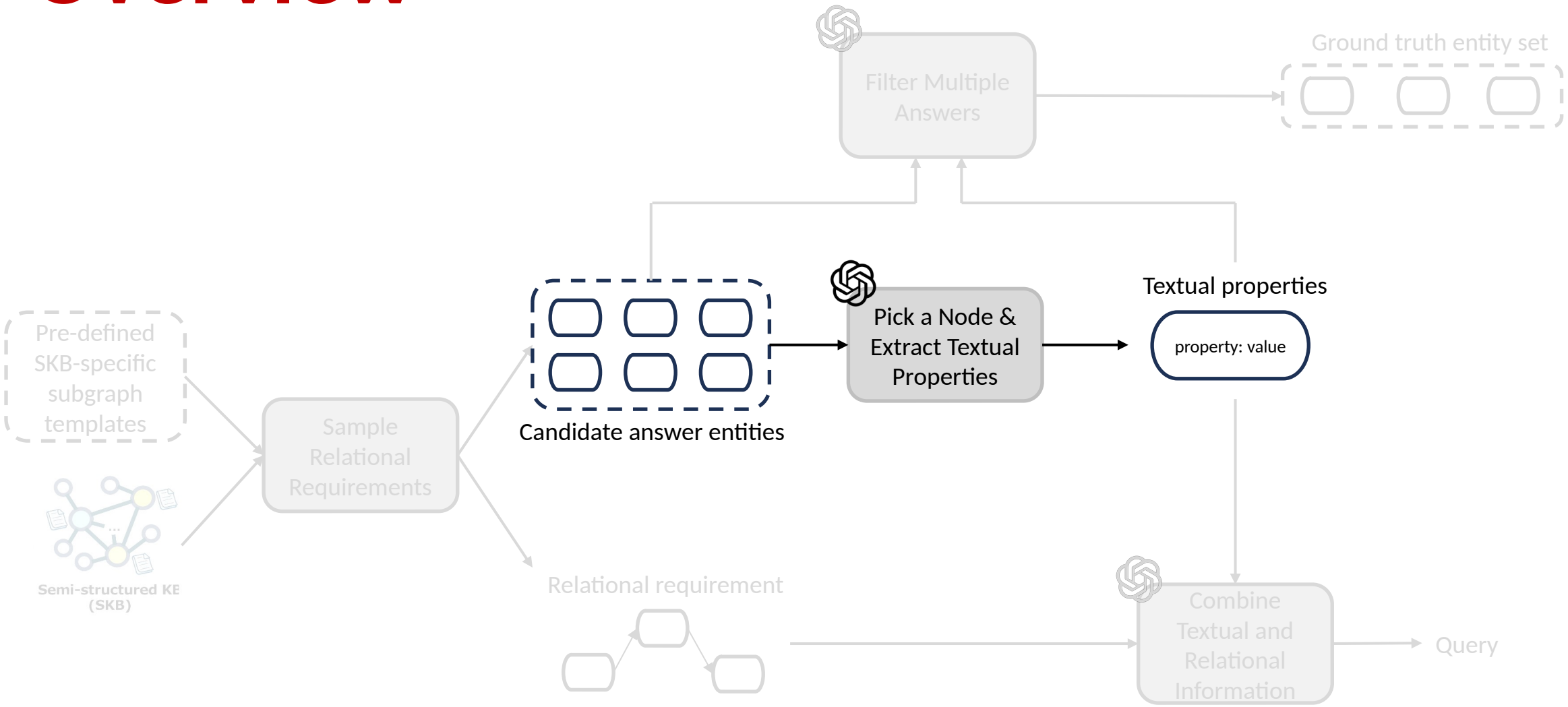
Query-Ground Truth Entity Set Generation

Overview



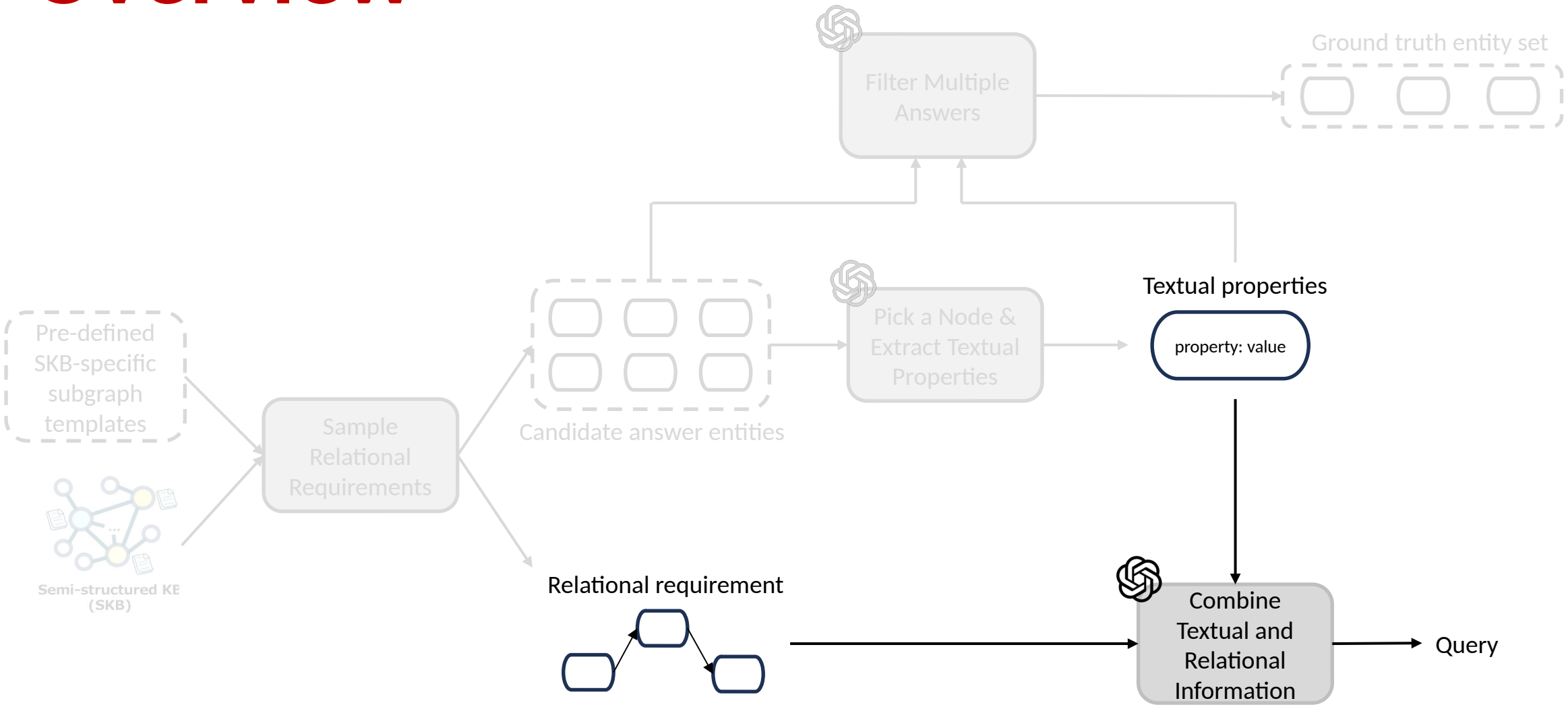
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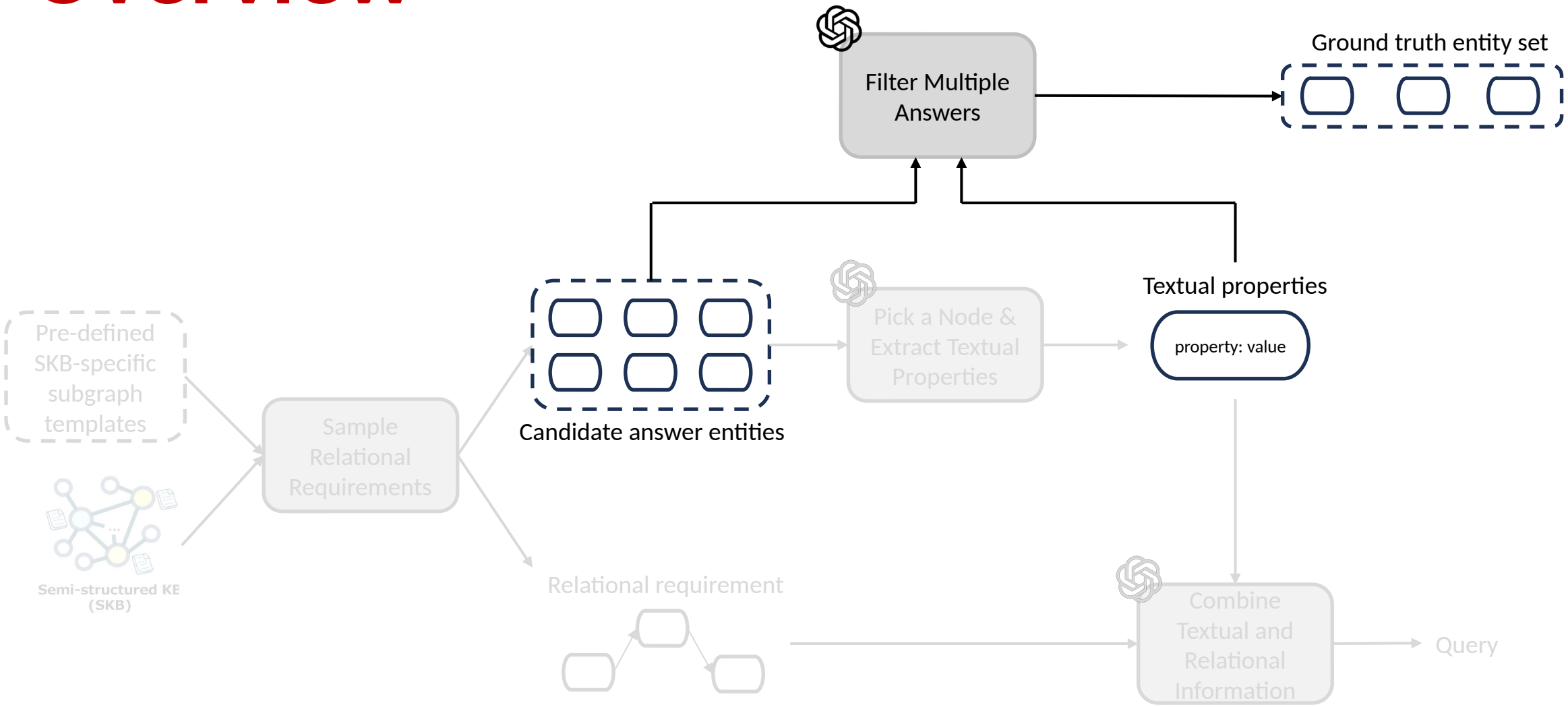
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Overview



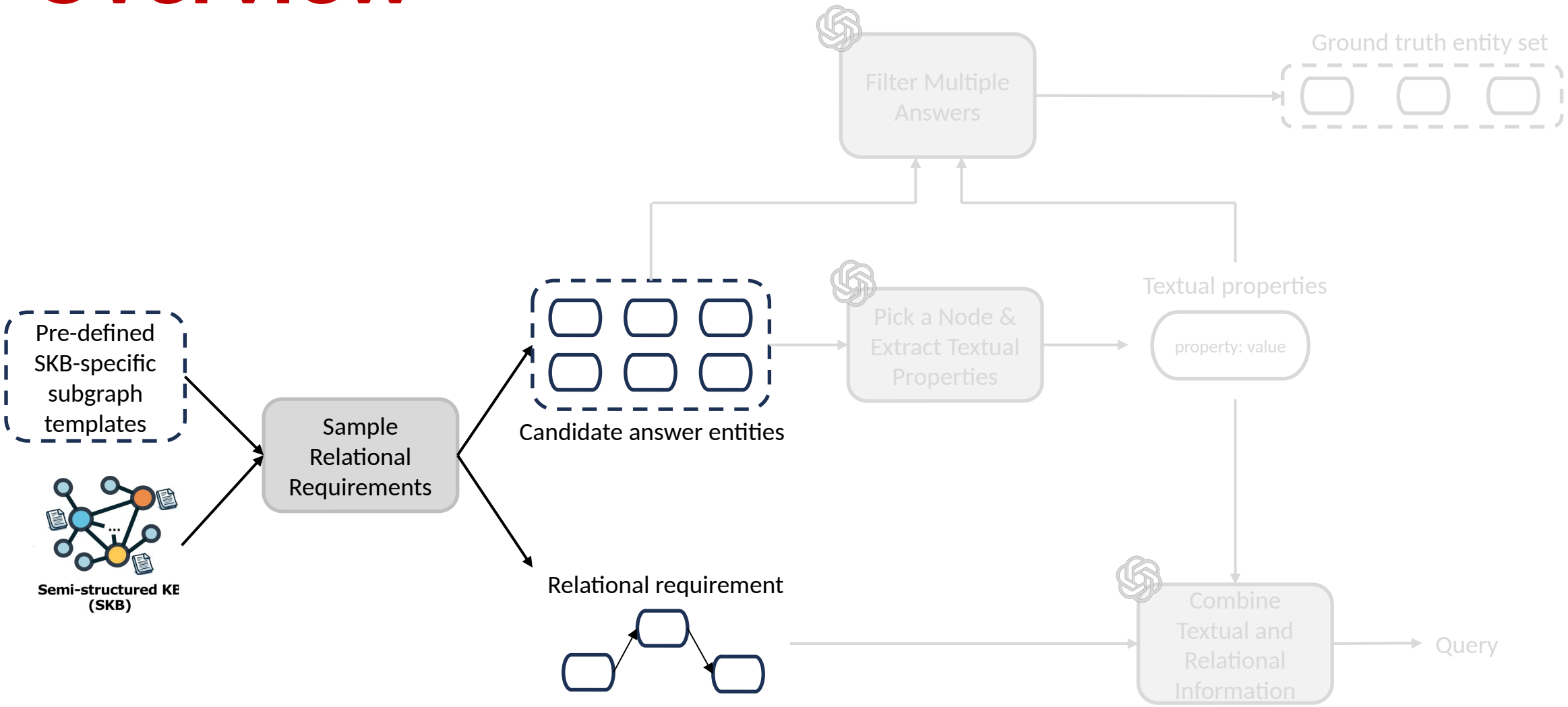
Query-Ground Truth Entity Set Generation

Overview



Query-Ground Truth Entity Set Generation

Overview



Query-Ground Truth Entity Set Generation

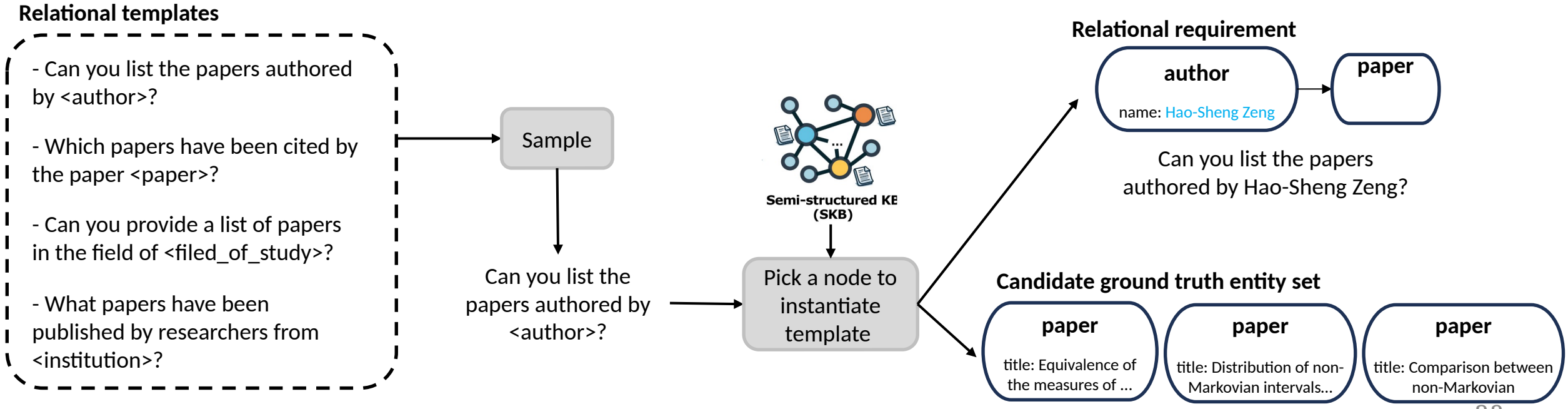
Step 1: Sample Relational Requirements

Initiate a relational requirement by sampling a relation template.

- **Relational requirement:** Subgraph expressed in string format
  - Nodes, node labels, relations, and one property-value is expressed.
- **Relation template**: A template for generating relational requirement

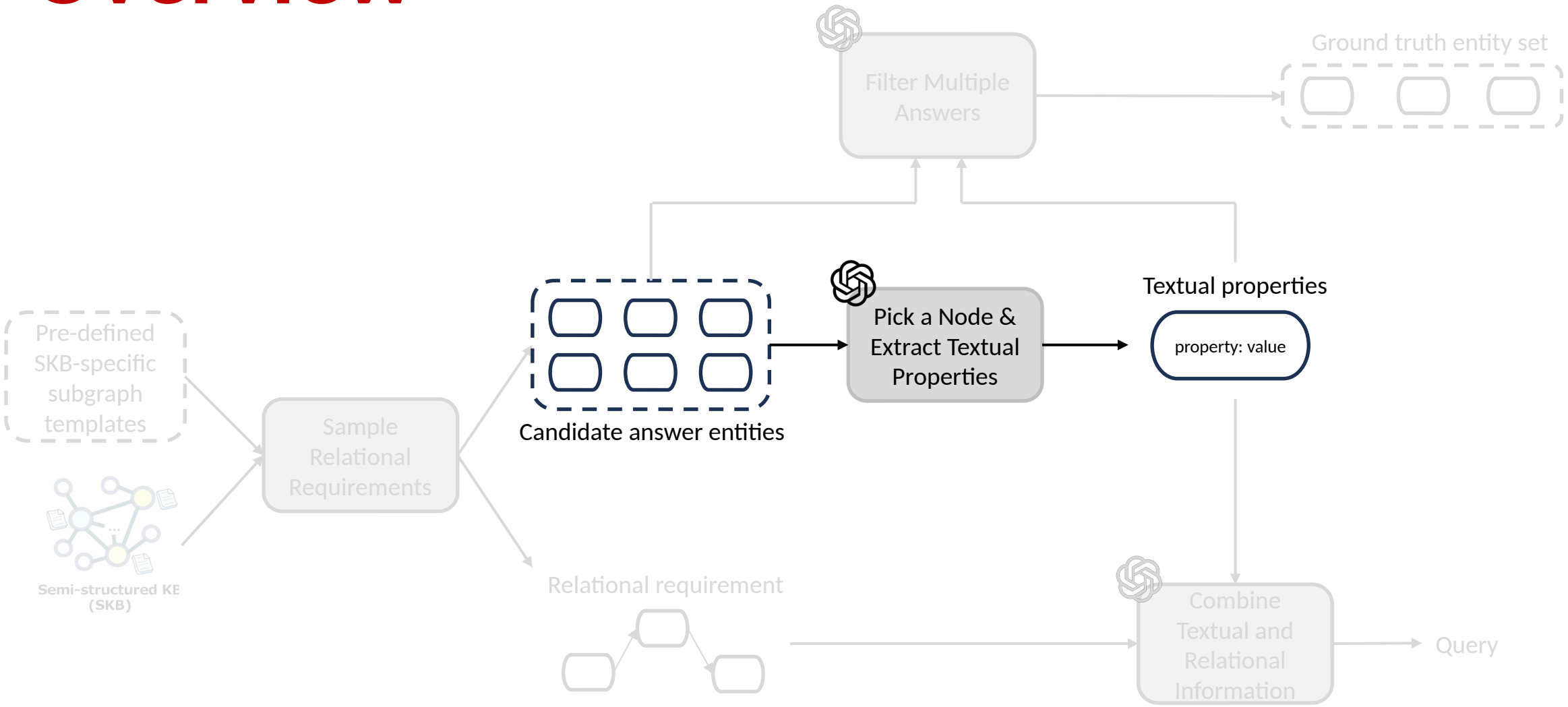
Global example
query: Search publications by Hao-Sheng Zeng on non-Markovian dynamics.
ground truth entity set:

- Distribution of non-Markovian intervals…
- Comparison between non-Markovian…



Query-Ground Truth Entity Set Generation

Overview



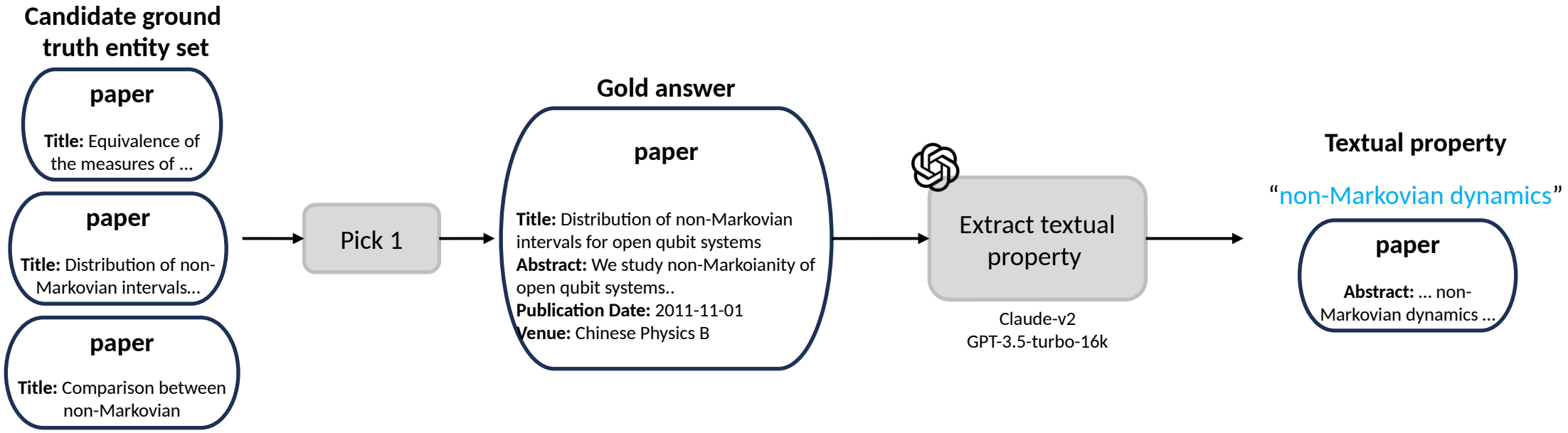
Query-Ground Truth Entity Set Generation

Step 2: Extract Textual Properties

1. Pick gold answer from candidate entity set.
2. Extract textual property from gold answer using LLM.
   - **Textual property:** Property-value constraint of gold answer expressed in string format.

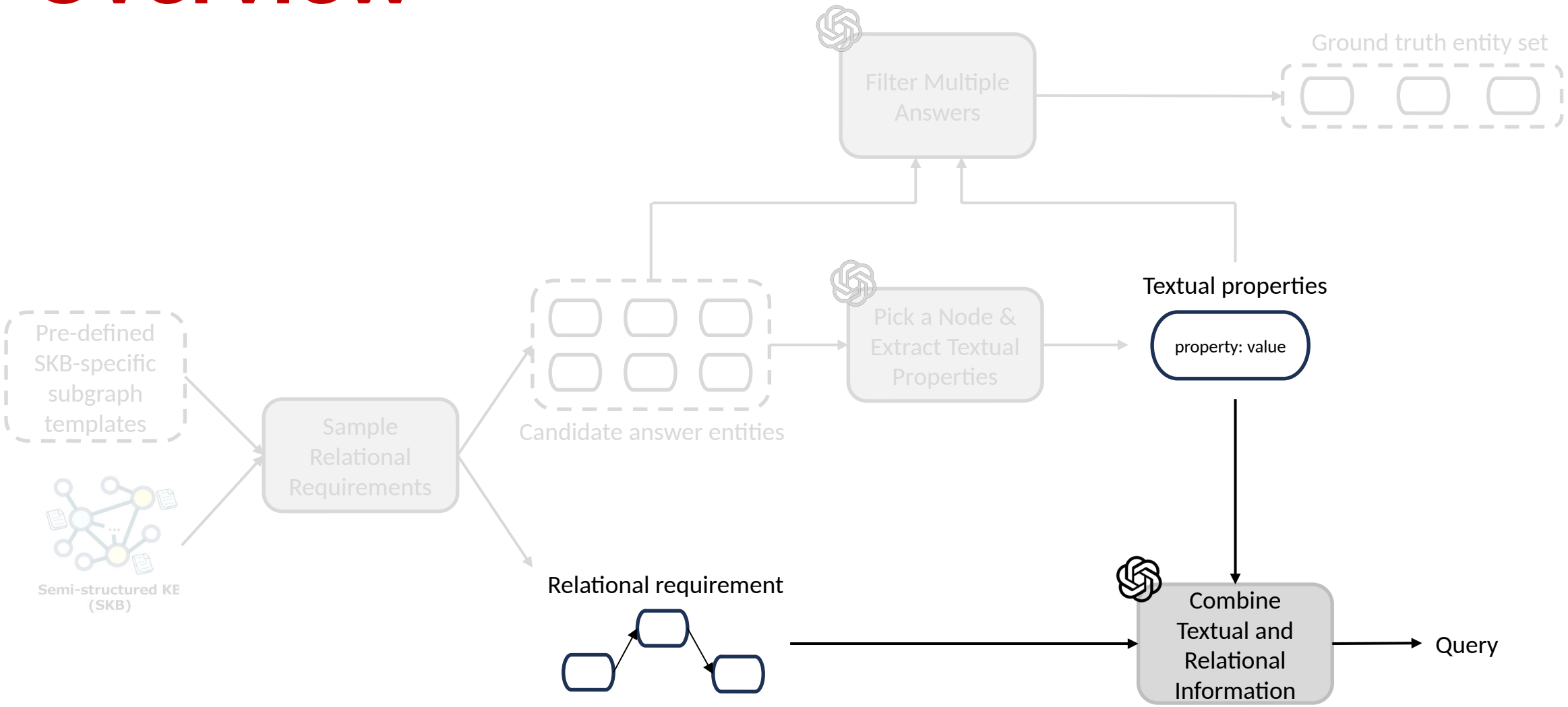
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Query-Ground Truth Entity Set Generation

Overview



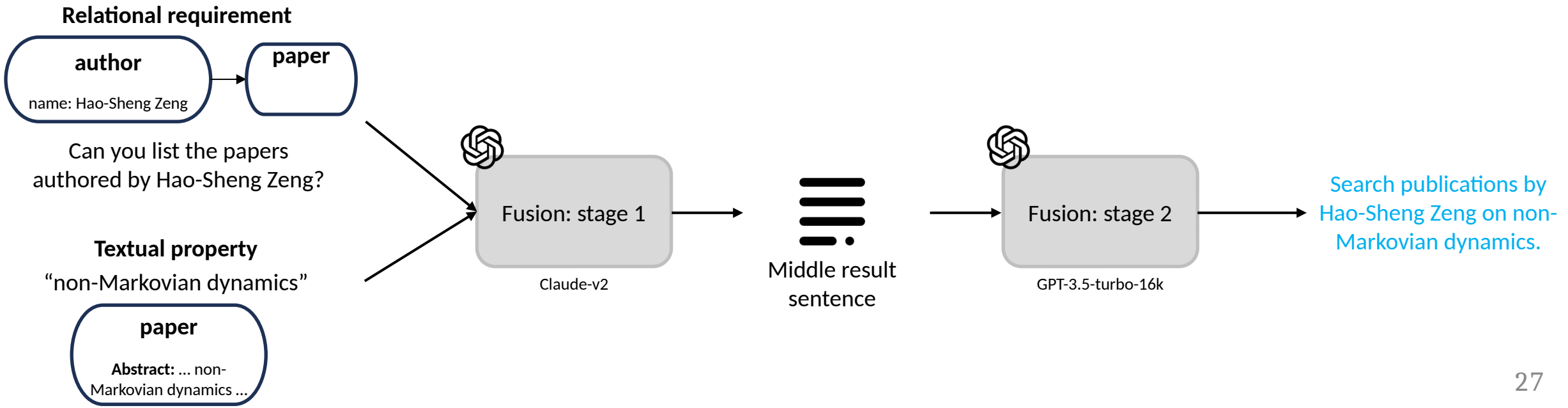
Query-Ground Truth Entity Set Generation

Step 3:Query Generation

Fuse relational requirement and textual property with LLMs.

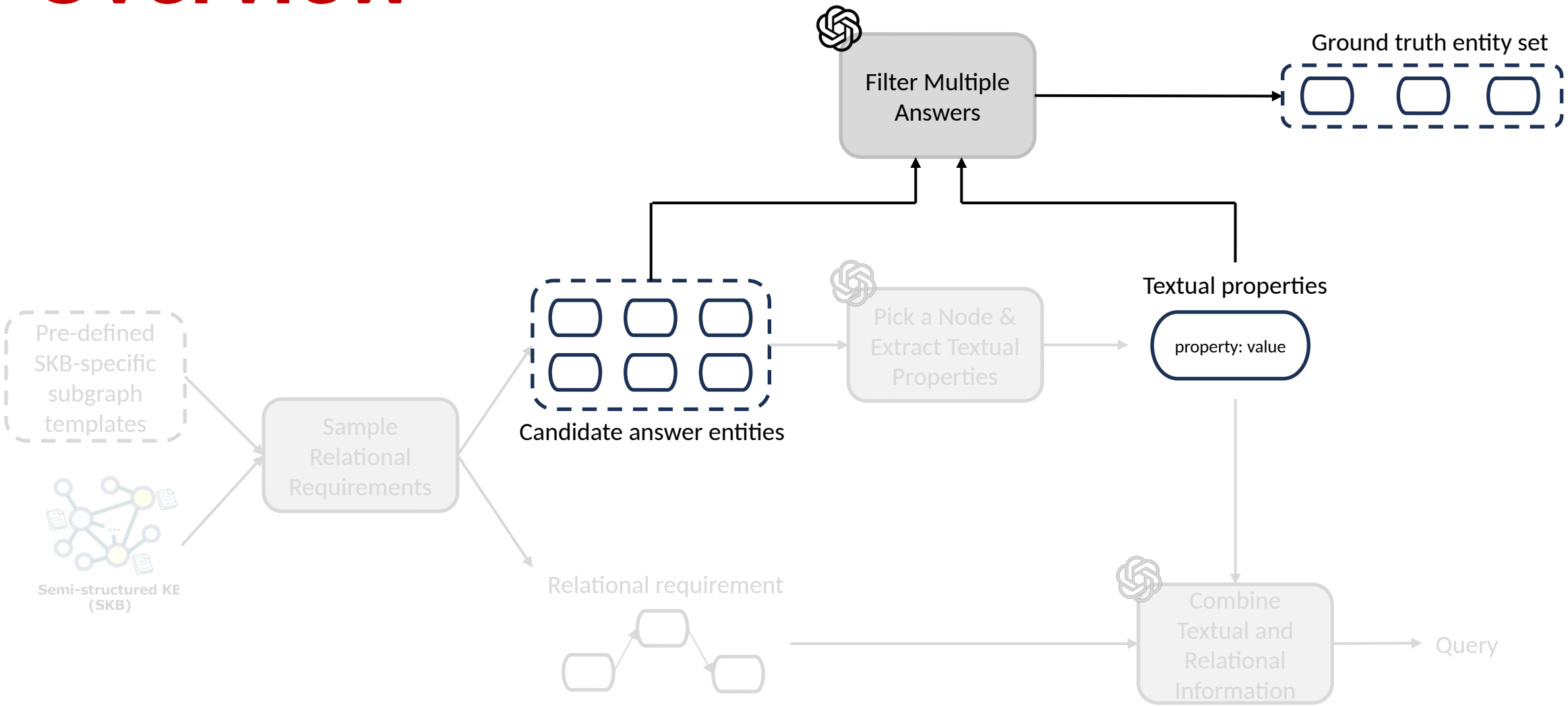
1. Ensure natural-sounding query
2. Enrich context, rephrase the language

Global example
query: Search publications by Hao-Sheng Zeng on non-Markovian dynamics.
ground truth entity set:
- Distribution of non-Markovian intervals…
- Comparison between non-Markovian…



Query-Ground Truth Entity Set Generation

Overview



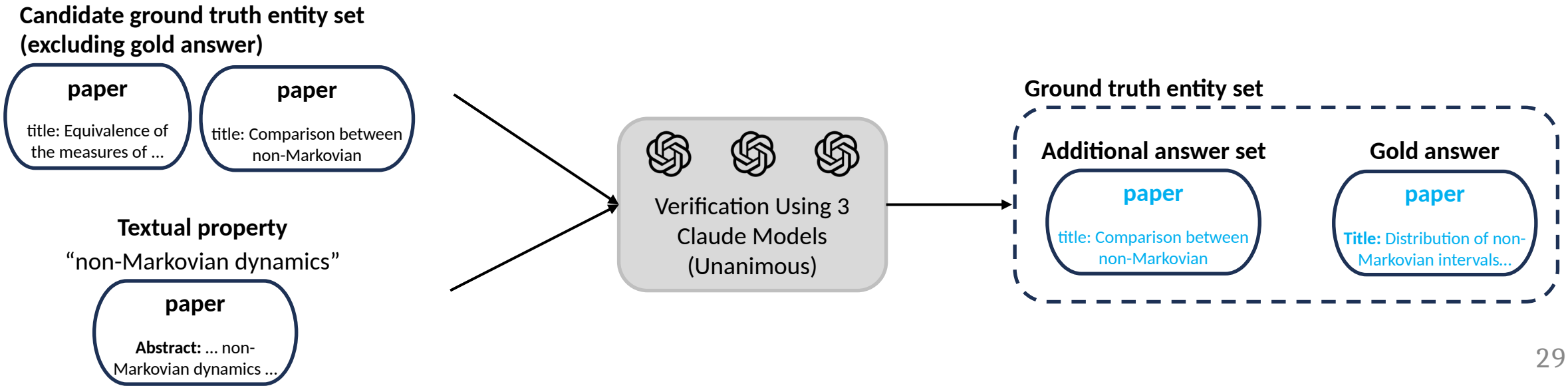
Query-Ground Truth Entity Set Generation

Step 4: Ground Truth Entity Set Generation

Assess whether each remaining candidate meets the additional textual requirement.

- Candidates that pass the verification across 3 Claude models are included in the final ground truth answer set.

Global example
query: Search publications by Hao-Sheng Zeng on non-Markovian dynamics.
ground truth entity set:
Distribution of non-Markovian intervals…
Comparison between non-Markovian…



Analysis

Query and Answer Length

- **Query length:** Number of tokens in a query
  - Reflects the amount of context information provided by user.
- **Answer length:** Number of ground truth answer nodes for a query
  - Indicates the level of inclusiveness or ambiguity of queries within a specific SKB.

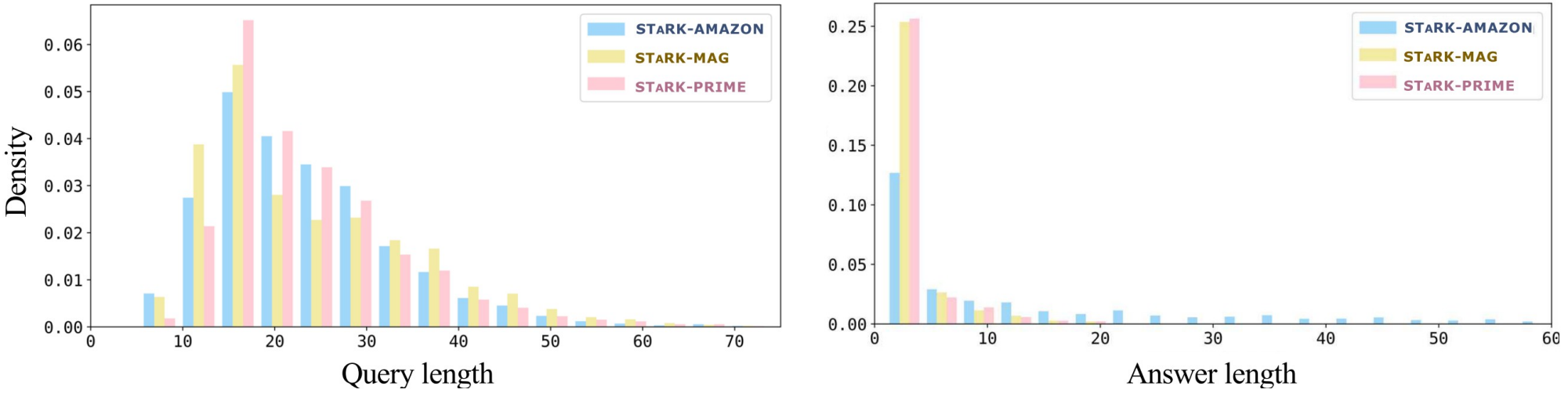


Figure 5. Distribution of query and answer lengths on STaRK datasets.

Query Diversity

- The diversity of query is measured with the diversity of tokens (token distribution).
- Entropy
  - $H(x) = -\sum_{x \in Vocab} p(x) \log p(x)$
- Type-Token Ratio
  - $TTR = \frac{\#\ of\ unique\ tokens}{Total\ number\ of\ tokens}$

	Shannon Entropy	Type-Token Ratio
STARK-AMAZON	10.39	0.179
STARK-MAG	10.25	0.180
STARK-PRIME	9.63	0.143
Reference[1]	10.44	0.261

Table 4. Query diversity measurement on STARK.

Ratio of Relational vs Textual Information

Analyze the proportionality between relational and textual information.

- Calculate the ratio of the length of relational requirements to the length of textual properties.

	Example query
STARK-AMAZON	*Looking for durable Dart World brand dart flights that resist easy tearing. Any recommendations?*
	What are recommended scuba diving weights for experienced divers that would fit well with my Gorilla PRO XL waterproof bag?
STARK-MAG	*Search publications by Hao-Sheng Zeng on non-Markovian dynamics.*
	What are some nanofluid heat transfer research papers published by scholars from Philadelphia University?
STARK-PRIME	*Could you provide a list of investigational drugs that interact with genes or proteins active in the epididymal region?*
	Search for diseases without known treatments and induce pruritus in pregnant women, potentially associated with Autoimmune.
	Please find pathways involving the POLR3D gene within nucleoplasm.
	Which gene or protein associated with lichen amyloidosis can bind interleukin-31 to activate the PI3K/AKT and MAPK pathways?

Table 5. Example pairs of query and answer node set for SKB retrieval task (Relational requirements are expressed in blue, and textual properties are expressed in green)

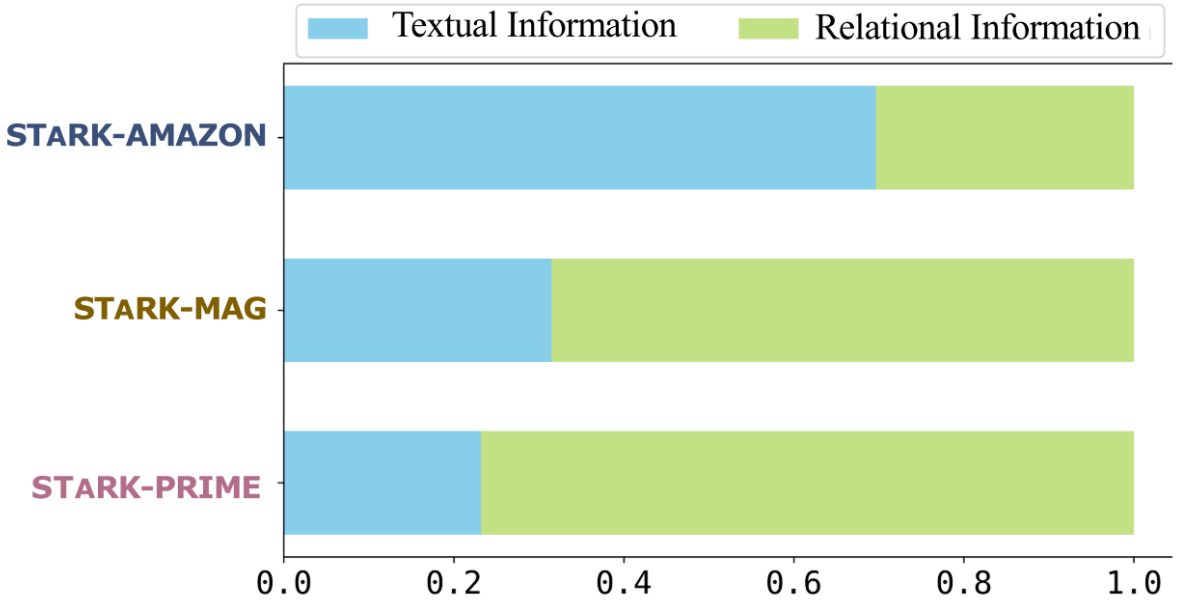


Figure 6. Average relative composition of relational requirement vs. textual property

Human Evaluation

63 participants evaluated STaRK with 3 aspects in 1 ~ 5 points:

- Natural-sounding (resembling natural conversation)
- Diverse (covering a wide range of question structures and complexity levels)
- Practical (relevant and useful in real-life situations)

	Naturalness	Diversity	Practicality
STARK-AMAZON	73.6 / 89.5	68.4 / 89.5	89.5 / 94.7
STARK-MAG	94.7 / 100	73.7 / 84.2	68.4 / 84.2
STARK-PRIME	67.8 / 92.8	71.4 / 82.1	71.4 / 89.3
Average	78.7 / 94.1	71.0 / 85.3	76.4 / 89.4

Table 6. Positive (4~) / Non-negative (3~) rates (%) of responses in each aspect of human evaluation.

Experiment

Experiment Goals and Settings

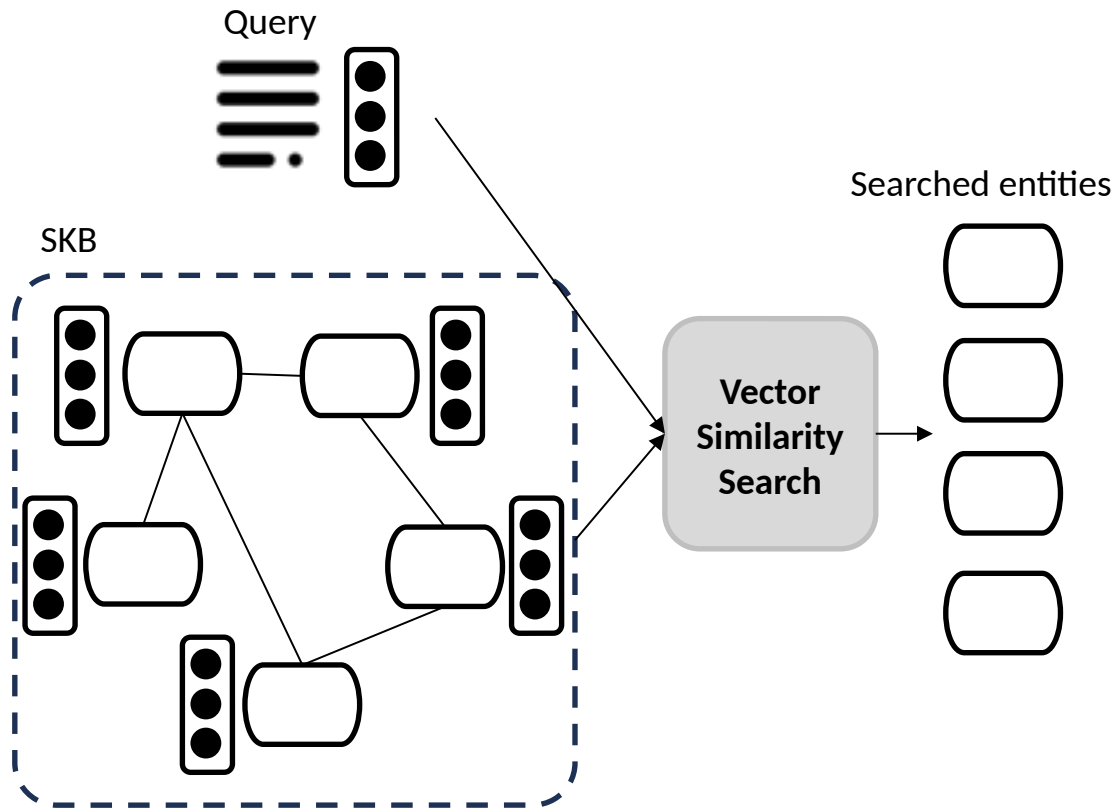
- **Goals**
  - How well do current retrieval approaches perform on our benchmark datasets?
  - What challenges are revealed by the experiments, and what are the potential future directions for improvement?

- **Setting**
  - GPU setting: NVIDIA A100-SXM4-80GB * 1

Proposed Baseline Retrieval Methods

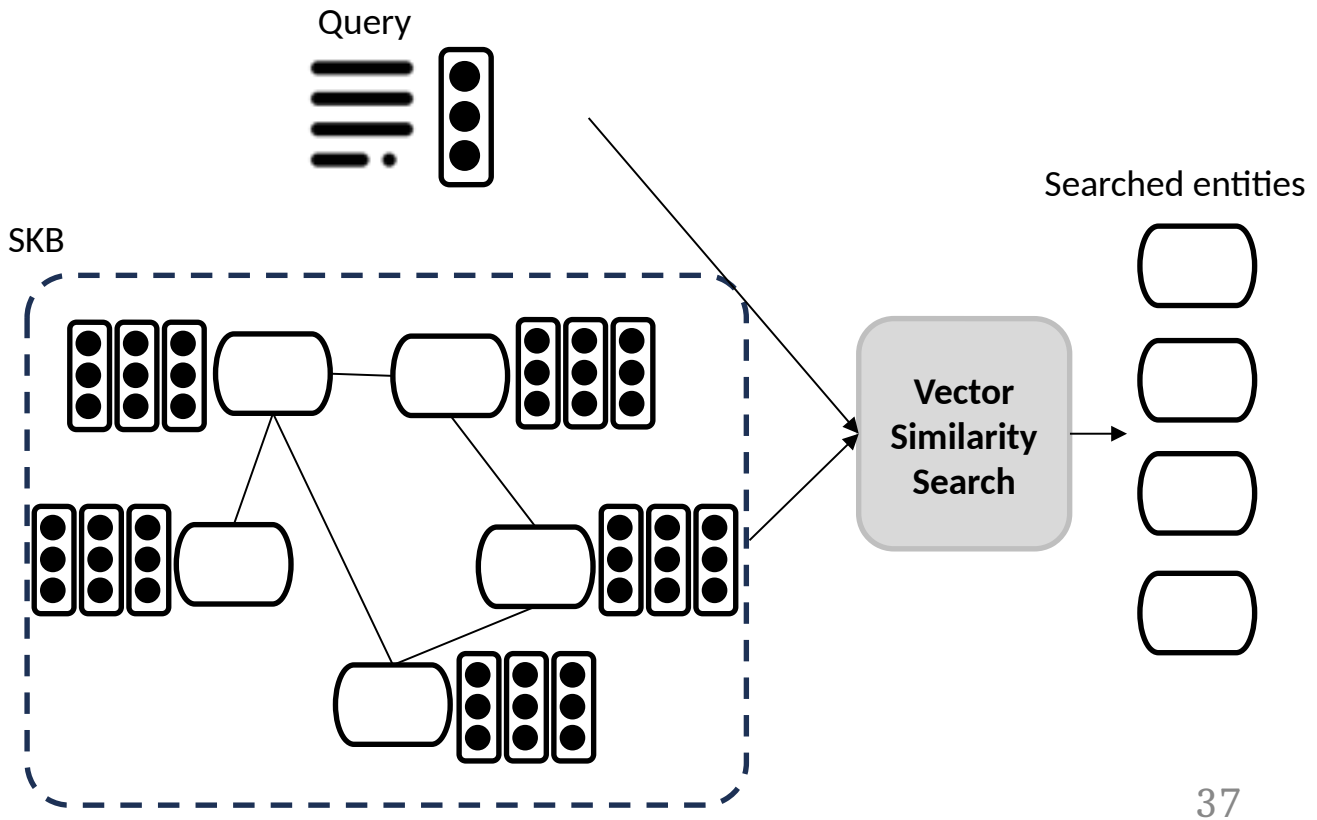
Vector Similarity Search

encoder = ada-002



Multi-Vector Similarity Search

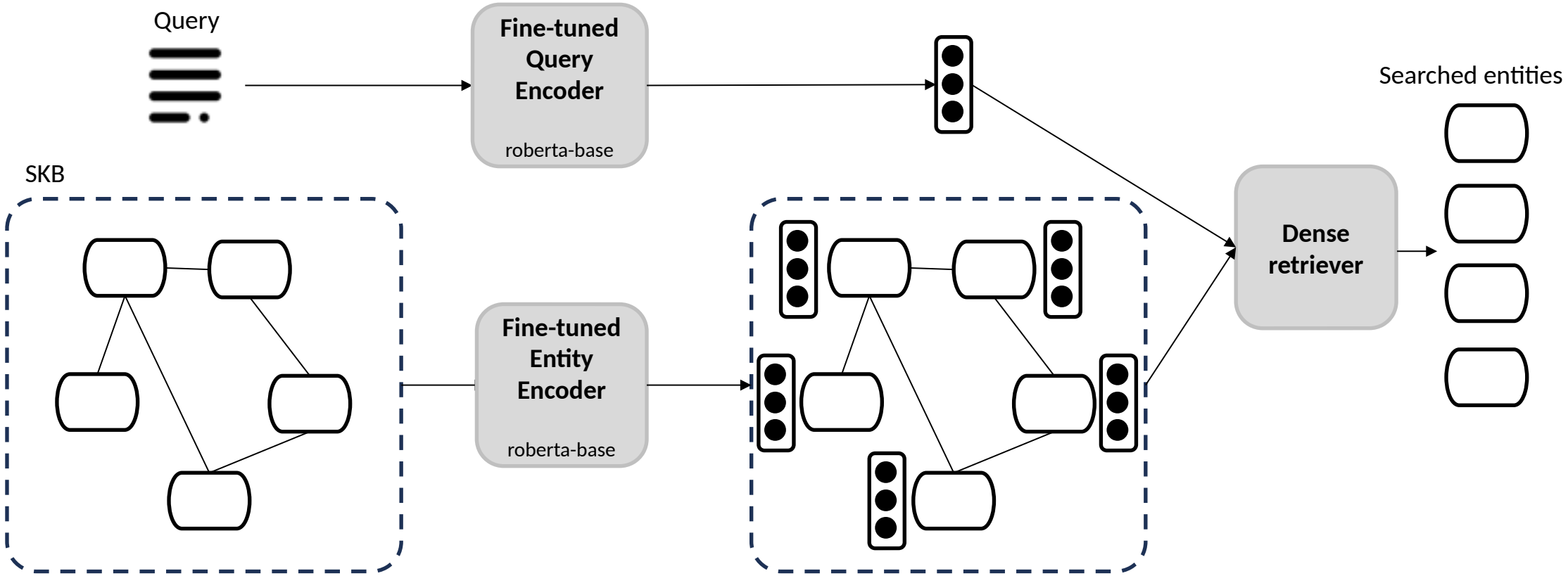
encoder = ada-002



Proposed Baseline Retrieval Methods

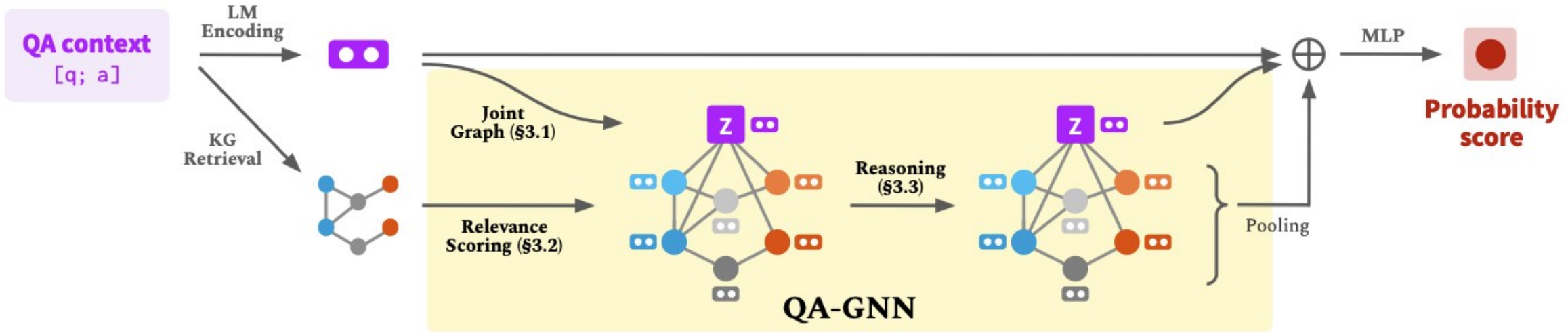
Dense Retriever

Encoder fine-tuning is done via contrastive learning, with 20 hard negatives per positive sample



Proposed Baseline Retrieval Methods

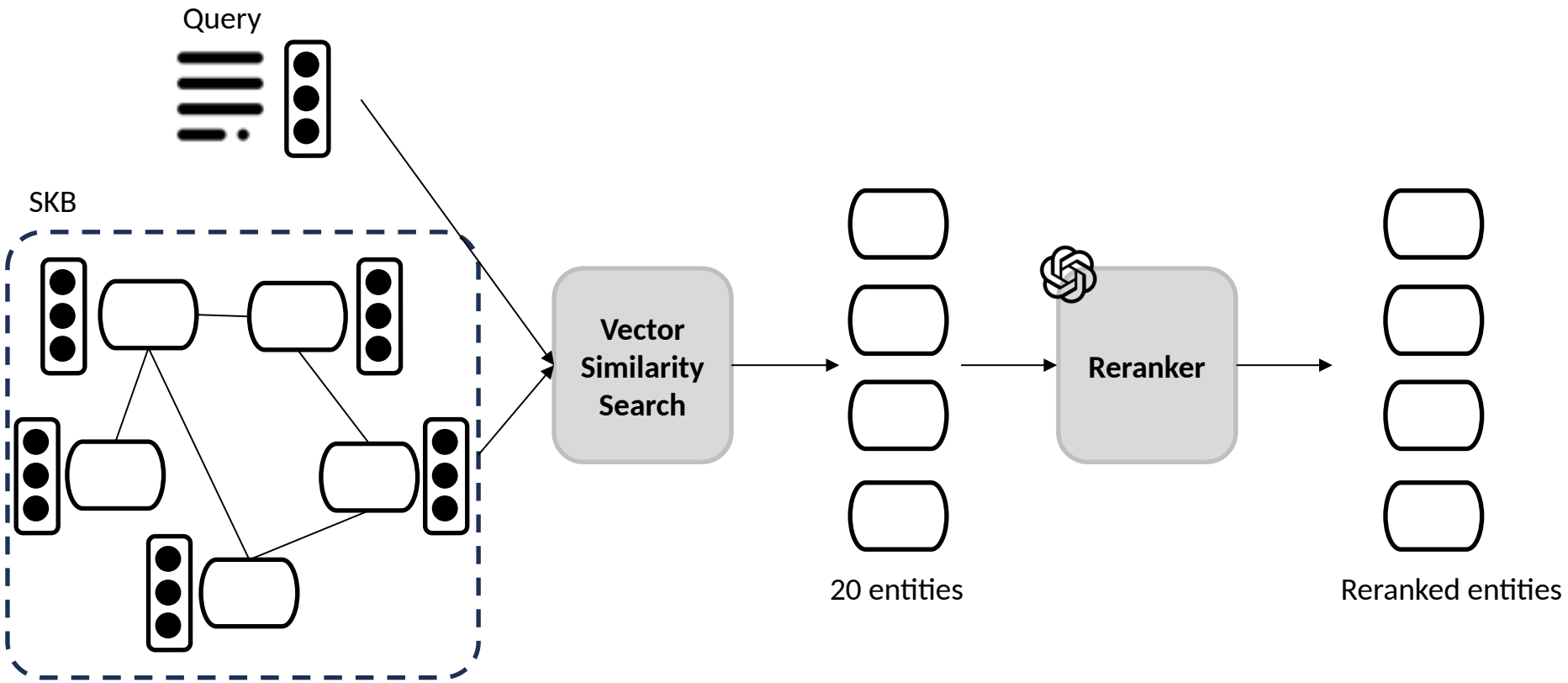
QAGNN



Proposed Baseline Retrieval Methods

VSS + LLM Reranker

encoder = ada-002



Metrics

Effectivity

- Hit@k
  - Hit@1
  - Hit@5
- Recall@k
  - Recall@20
- Mean Reciprocal Rank

Efficiency

- Latency

Query: Could you provide a list of investigational drugs that interact with genes or proteins active in the epididymal region?

Ground truth entities

(S)-3-phenyllactic Acid | Anisomycin | Puromycin

Search results (sorted in descending similarity scores):

1. Apimostinel
2. **Anisomycin**
3. Arketamine
4. Enobosarm
5. **Puromycin**
6. Endoxifen
7. Leflutrozole
8. **(S)-3-phenyllactic Acid**
9. MVT-602
10. HCP-1302

Hit@1	Hit@5	Recall@5	MRR
0	1	2/3	1/2

Figure 7. An example search result and its evaluation.

Main Experimental Results

		Dense Retriever (roberta)	QAGNN (roberta)	VSS (ada-002)	Multi-VSS (ada-002)	VSS+Claude2 Reranker	VSS+GPT4 Reranker
STARK-AMAZON	Hit@1	0.1529	0.2656	0.3916	0.4007	0.4266	**0.4479**
	Hit@5	0.4793	0.5001	0.6273	0.6498	0.6746	**0.7117**
	Recall@20	0.4449	0.5205	0.5329	0.5512	0.5376	**0.5535**
	MRR	0.3020	0.3775	0.5035	0.5155	0.5329	**0.5569**
STARK-MAG	Hit@1	0.1051	0.1288	0.2908	0.2592	0.3202	**0.4090**
	Hit@5	0.3523	0.3901	0.4961	0.5043	0.5334	**0.5818**
	Recall@20	0.4211	0.4697	0.4836	**0.5080**	0.4834	0.4860
	MRR	0.2134	0.2912	0.3862	0.3694	0.4129	**0.4900**
STARK-PRIME	Hit@1	0.0446	0.0885	0.1263	0.1510	0.1611	**0.1828**
	Hit@5	0.2185	0.2135	0.3149	0.3356	0.3582	**0.3728**
	Recall@20	0.3013	0.2963	0.3600	**0.3805**	0.3598	0.3405
	MRR	0.1238	0.1473	0.2141	0.2349	0.2466	**0.2655**

Table 7. Main experimental results.

Efficiency Experiment Result

	Dense Retriever	QAGNN	VSS	Multi-VSS	VSS+Claude	VSS+GPT4
STARK-AMAZON	2.34	2.32	5.71	4.87	27.24	24.76
STARK-MAG	0.94	1.35	2.25	3.14	22.60	23.43
STARK-PRIME	0.92	1.29	0.54	0.90	29.14	26.97
Average	1.40	1.65	2.83	2.97	26.33	25.05

Table 8. Latency (s) of the retrieval systems on STaRK.

APPENDIX

Generated SKB 1

Amazon Semi-structured Knowledge Base

Generation method:

- Generation method of knowledge graph is not denoted.
- Textual information for products obtained by
  - the meta data from *Amazon Product Reviews*.
  - customer Q&A records from *Amazon Question and Answer Data*.

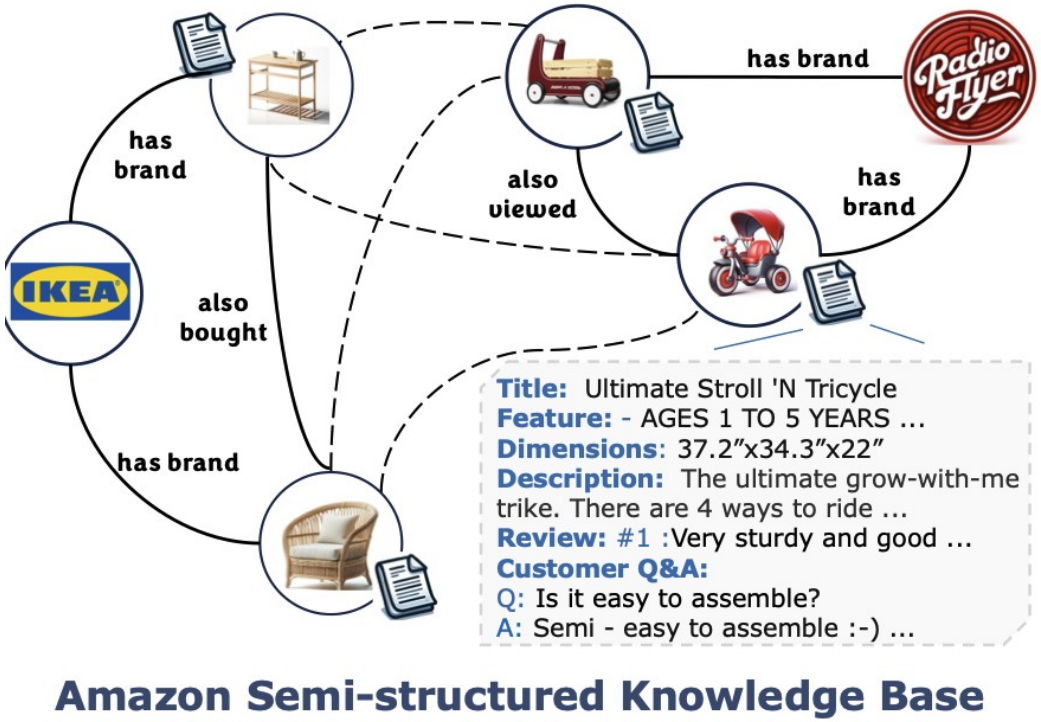


Figure 4. The Amazon semi-structured knowledge base

Generated SKB 3

Prime Semi-structured Knowledge Base

Generation steps:

1. Leverage the existing knowledge graph *PrimeKG*
2. Integrate the textual details from multiple databases for gene/protein and pathway entities such as genomic position, gene activity summary and pathway orthologous event.

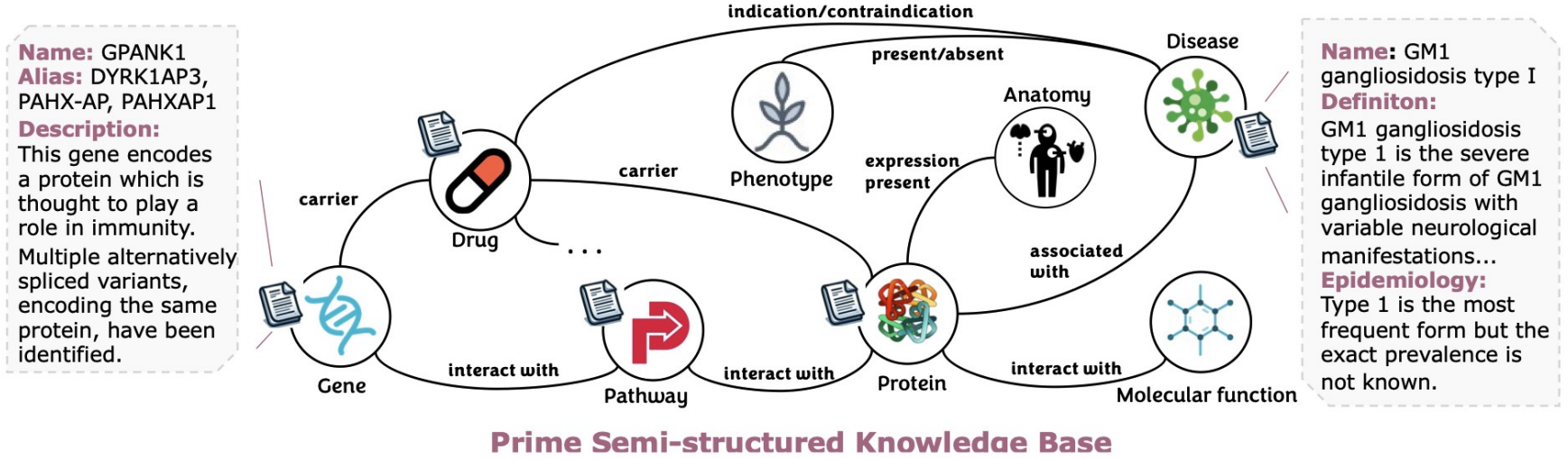


Figure 6. The Prime semi-structured knowledge base